

LESSON GUIDE

Technology Plus

Unit 7: Security

6.1.2 – Integrity Concerns

Lesson Overview:

In this lesson, students will assess integrity concerns within the CIA Triad and explore how data can be validated to detect unauthorized changes and ensure its accuracy and trustworthiness.

Students will:

- Define integrity as it relates to the CIA Triad.
- Identify common threats to integrity.
- Examine best practices for maintaining and verifying data integrity.

Guiding Question: What are some ways data integrity can be compromised, and how can you tell if your data has been altered without your knowledge?

Suggested Grade Levels: 8-12

Technology Needed: Yes

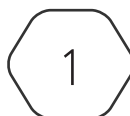
Approximate Time Required: 55-65 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.1 Explain fundamental security concepts and frameworks.
 - Confidentiality, **integrity**, and availability

This content is based upon work supported by the US Department of Homeland Security's Cybersecurity & Infrastructure Security Agency under the Cybersecurity Education Training and Assistance Program (CETAP).



CYBER.ORG
THE ACADEMIC INITIATIVE OF THE CYBER INNOVATION CENTER

Copyright © 2025 Cyber Innovation Center
All Rights Reserved. Not for Distribution.

Lesson Background

Background Information:

As one of the core principles of the CIA Triad, maintaining the integrity of information is crucial for ensuring the trustworthiness and accuracy of data. Integrity ensures the data is reliable and free from unauthorized modifications. Threats to integrity involve data tampering which mostly occurs with data at rest or in use. Data in motion can be modified through various methods such as replay and on-path attacks. Implementing best practices such as using file hashing and digital signatures to check for file integrity as well as using up to date security protocols with HTTPS and WPA2/WPA3 for Wi-Fi can help to mitigate these threats.

Materials Needed:

Materials per Class	• Lesson Guide 6.1.2 – Integrity Concerns • Lesson Slides 6.1.2 – Integrity Concerns • Lab Slides 6.1.2 – Hashing for Integrity
Materials per Student	• Student Handout 6.1.2 – Integrity Concerns • Student Notes 6.1.2 – Integrity Concerns • Student Assessment 6.1.2 – Integrity Concerns

Cyber Terms:

Term	Definition
data tampering	unauthorized modification, removal, or manipulation of data
replay attack	when data is captured in transit on a network, usually between a client and a server, to be used later to mimic an authorized transaction
on-path attack	also called adversary in the middle attack, formerly known as man-in-the-middle attack, this captures data in transit, usually between a client and server, to establish an exposed connection between a user and can be exploited
file hashing	using a cryptographic algorithm to generate a string of characters based on the contents of a file to produce a unique signature known as a checksum
checksum	the resulting value after running a hash function on a file or on a set of data which serves as a digital signature to verify file integrity
digital signature	used to authenticate the integrity of documents or files using a cryptographic key where the sender uses a private key, and the recipient uses a public key to verify the integrity

6.1.2 - Integrity Concerns

Guiding Question: What are some ways data integrity can be compromised, and how can you tell if your data has been altered without your knowledge?

Lesson Launch: (3-5 minutes)

- Present the scenario that the school's student information system (SIS) has been breached, and grades have been switched with other students' grades.
- Have students discuss:
 - How would this impact students?
 - How would this impact teachers?
 - How would this impact the school as a whole?

Keeping Integrity Real (25-30 minutes)

Review of the CIA Triad and Integrity

- Distribute Student Handout 6.1.2 – Integrity Concerns and encourage them to fill in answers as you go through today's lesson.
- Remind students that the CIA Triad focuses on confidentiality, keeping data secure from unauthorized viewers, integrity, keeping data trustworthy, and a availability, keeping it accessible.
- Integrity centers around keeping data accurate and reliable. This means that the data has not been altered by an unauthorized user or in a way that was unintended such as accidental editing.
- Examples of data you would not want altered:
 - Medical records
 - Financial records
 - Legal records
 - Educational records

Threats to Integrity – Data Tampering

- **Data tampering**; Unauthorized modification, removal, or manipulation of data.
- This could be caused by attackers, insider threats such as a disgruntled employee, accidents, or impersonations of authorized users.

Threats to Integrity – Replay Attacks

- **Replay attacks** occur when an attacker captures data in transit using network snooping software or hardware, usually between a client and server, which is then used later.

- Emphasize that this could also be considered a breach of confidentiality if sensitive data is captured; but replay attacks are primarily integrity violations—because the attacker reuses previously validated data to fraudulently perform actions the original sender never intended.

Threats to Integrity – On-Path Attacks

- Also known as adversary, or man-in-the-middle attacks, **on-path attacks** capture data in transit, usually between a client and server, to establish a live connection that can be exploited.
- These are like replay attacks; however, instead of simply capturing the data for use later, the attacker alters the data while in transit, so all the authentication data appears legitimate.
- These attacks require software or hardware to act as a mediator between the client and service they are attempting to use.
- It's important to note that while both attacks involve stolen authentication data—like credentials, tokens, or session cookies—replay attacks are passive and reuse that data to repeat a valid transaction, whereas on-path attacks are active and alter the outcome in real time.

Ensuring Integrity – Data Tampering

- **File hashing** takes the contents of the file, whether it be a document or application file, and uses a cryptographic algorithm to generate a unique string of characters. Comparing this string with known originals can tell you whether alterations were made.
- **Digital signatures** can also be generated for files, emails, and more. These act like digital stamps. The sender, using a private cryptographic key, and receiver, using a public key, can verify the validity of the communication or data sent between.

Ensuring Integrity – Replay and On-Path Attacks

- Using only secure protocols like https or wireless networks using WPA2 or WPA3 can help mitigate these attacks.
- If a public Wi-Fi network must be used, do not perform transactions you would consider secure or sensitive.
- Digital signatures can also be used for communication to verify messages were not altered in transit.

Lab: 6.1.2 – Hashing for Integrity (30-35 minutes)

- This lab will use checksums to verify the integrity of files.
- Students will use the MD5 and SHA512 checksums to validate files and check for integrity. Students will then copy a configuration file, make edits, and observe the changes in the resulting hash values. They will end the lab with using the diff command to observe changes between multiple files.
- It is suggested that login information be distributed and virtual machines launched prior to the start of the lab.
- What are checksums and how can they be used to check for file integrity?

A checksum is a unique string of characters created by hashing a file with a cryptographic algorithm. If the file changes in any way, the checksum will also change, so comparing checksums can show whether the file has been altered.

- What is a reason you may want to check system configuration files such as the `sshd_config` file?
Because configuration files control how systems behave. If someone changes a setting, like disabling password rules or changing who can log in, it could create security risks. Checking the file's integrity helps catch that.
- What does the `diff` command allow you to do?
It shows the differences between two text files, line by line. This makes it easier to find what was added, removed, or changed, especially in important files like configurations or code.

Putting It All Together (3-5 minutes)

Discussion Questions

- What are some examples of data tampering and how can you validate the integrity of data or files?
Examples of data tampering include unauthorized edits to system files, a disgruntled employee changing payroll data, or someone spoofing an identity to alter grades. To check for tampering, you can use tools like file hashing or digital signatures. If the hash or signature doesn't match the original, the file has been changed.
- How does a replay attack differ from an on-path attack?
A replay attack is passive; it captures valid data (like credentials or tokens) during a real session and uses it later to fake a new transaction. An on-path attack is active and happens in real time; the attacker intercepts and modifies data while it's being transmitted between systems.
- What are some ways to protect yourself from network exploits like that of a replay attack?
Use secure protocols like HTTPS, avoid public Wi-Fi when handling sensitive information, and make sure you're using secure Wi-Fi standards like WPA2 or WPA3. Digital signatures can also help detect if transmitted data has been tampered with.